Example:

1. A free particle moving along x-axis needs only one coordinate to specify it completely. So its degree of freedom is one.
2. Similarly a particle moving over a plane has two degrees of freedom.
3. A particle moving in space has three degrees of freedom.

Suppose if we have N number of gas molecules in the container, then the total number of degrees of freedom is f = 3N.

But, if the system has *q* number of constraints (restrictions in motion) then the degrees of freedom decreases and it is equal to $f = 3N\text{-}q$ where N is the number of particles.

9.3.2 Monoatomic molecule

A monoatomic molecule by virtue of its nature has only three translational degrees of freedom.

Therefore $f = 3$

Example: Helium, Neon, Argon

9.3.3 Diatomic molecule

There are two cases.

1. At Normal temperature

A molecule of a diatomic gas consists of two atoms bound to each other by a force of attraction. Physically the molecule can be regarded as a system of two point masses fixed at the ends of a massless elastic spring.

The center of mass lies in the center of the diatomic molecule. So, the motion of the center of mass requires three translational degrees of freedom (figure 9.5 a). In addition, the diatomic molecule can rotate

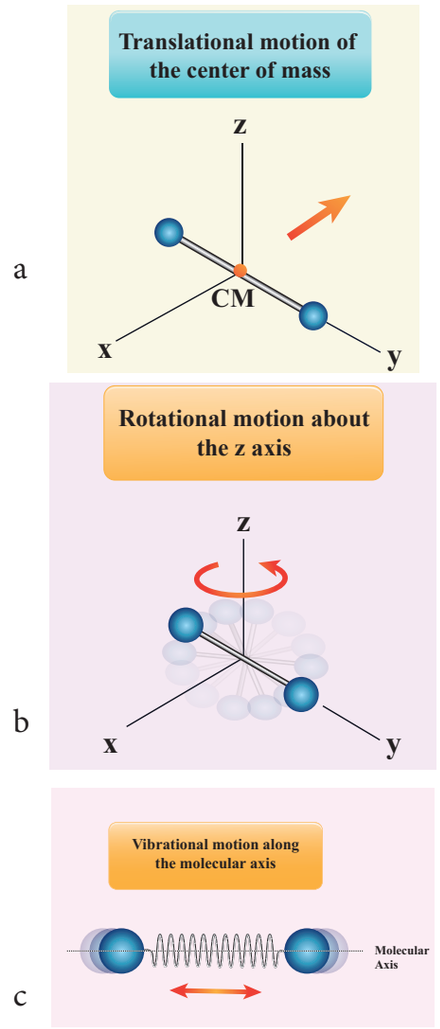


Figure 9.5 Degree of freedom of diatomic molecule

about three mutually perpendicular axes (figure 9.5 b). But the moment of inertia about its own axis of rotation is negligible (about y axis in the figure 9.5). Therefore, it has only two rotational degrees of freedom (one rotation is about Z axis and another rotation is about X axis). Therefore totally there are five degrees of freedom.

$$f = 5$$

2. At High Temperature

At a very high temperature such as 5000 K, the diatomic molecules possess additional

UNIT-9(XI-Physics_Vol-2).indd 174 17-08-2020 16:21:31